Vaibhav Singh

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Backend Software Engineer (SDE-1) with **2+ years** of experience designing and building **scalable, high-performance microservices and REST APIs** using **Java and Spring Boot**. Delivered 0-to-1 backend systems at Cleartrip (Flipkart Group) and optimized high-throughput data pipelines at ZL Technologies. Skilled in **system design (HLD/LLD), distributed systems, MySQL, Elasticsearch, Aerospike caching, Kafka and performance tuning**, with a strong focus on reliability, scalability and clean API design.

EXPERIENCE

Software Development Engineer I (Backend) Feb 2026 – Present
Cleartrip (Flipkart Group) *Bangalore, India*

- Owned **0-to-1 backend development** of Cleartrip's Holiday Packages platform, building **10+ production REST APIs** in **Java and Spring Boot microservices** for search, listing, pricing, filters and booking.
- Designed **HLD and LLD** for the packages platform, defining service boundaries, API contracts and data flow across microservices.
- Engineered **Elasticsearch**-based search, filter and homepage feed APIs using optimized queries and aggregations, enabling fast package discovery with **low-latency (<150ms)** responses.
- Reduced repeated database reads for high-traffic user APIs by implementing **Aerospike caching** with cache eviction on data updates.
- Integrated third-party supplier APIs for live pricing with **circuit breakers, timeouts and fallbacks**, ensuring **high availability and fault tolerance** when external services slow down.
- Developed an **event-driven lead management service** that processes real-time user activity webhooks with lead scoring, de-duplication and **Salesforce** integration for the sales team.
- Improved **observability** and production debugging by introducing **Kafka and Avro** API event logging across all package microservices.
- Eliminated a **production startup crash across 4 microservices** by tracing an Aerospike client timeout issue and making it configurable.
- Enabled business teams to publish **30+ packages** without deployment by migrating package content to a **database-backed, auto-refreshing catalog**.
- Partnered with **frontend, QA, product and infra teams** to drive end-to-end API integration, resolve reported bugs, and manage deployments and service access across environments.

Software Developer Sep 2024 – Jan 2026
ZL Technologies *Hyderabad, India*

- Redesigned I/O-bound data pipelines processing **millions of log events/day** — optimized batching and resource pooling — lifting end-to-end production throughput by **30%**.
- Resolved lock contention in multithreaded aggregation pipelines by migrating to **lock-free concurrent data structures**, reducing enterprise dashboard staleness from batch-level delays to **under 1 minute**.
- Delivered **3 consecutive sprint milestones** without scope slippage while concurrently code-reviewing and task-scoping work for 2 junior engineers across critical production modules.
- Collaborated with cross-functional teams (product, frontend, QA) in an **Agile/Scrum environment** to deliver features efficiently.

PROJECTS

API Logging & Metrics Aggregation System
ZL Technologies *Java, Log4j2, Apache ORC, SQL*

- Cut analytics storage by **50%** and converted overnight batch runs into **1-minute** rolling refreshes by migrating high-volume API logs from raw files to **Apache ORC** — giving dashboards near-real-time production data.
- Architected a zero-downtime Graph API interception layer that processed **10k+ requests/hour** at **<5 ms** added latency, increasing observability coverage by **40%** across enterprise services.
- Implemented SQL aggregations for **latency percentiles, TPS, and error rate** across millions of daily events — giving engineering teams real-time, production-grade visibility into service health KPIs.

SKILLS

- **Languages:** Java, SQL, JavaScript, C++
- **Backend:** Spring Boot, REST APIs, Microservices, Distributed Systems, System Design (HLD/LLD), Event-Driven Architecture, Multithreading, Concurrency, OOP, Design Patterns, Caching, Circuit Breakers, Fault Tolerance, Scalability, Observability, Unit Testing
- **Databases, Search & Messaging:** MySQL, PostgreSQL, MongoDB, Cassandra, MS SQL Server, Aerospike, Elasticsearch, Apache Kafka, Avro, Apache Spark, Apache ORC, Query Optimization, Indexing
- **Cloud, DevOps & Tools:** GCP (Cloud Run, GCS), Docker, CI/CD, Gradle, Maven, Git, IntelliJ IDEA, Postman, Cursor, Claude Code, Agile/Scrum
- **Core CS:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

EDUCATION

National Institute of Technology, Kurukshetra Dec 2020 – May 2024
B.Tech, Electronics and Communication Engineering | *CGPA: **8.66/10*** *Haryana, India*

ACHIEVEMENTS

- **CodeChef 4-Star** (max rating **1705**); solved **700+** problems across LeetCode, CodeChef and Codeforces (graphs, DP, advanced data structures).
- Department **rank holder, Semester 1**, ECE, NIT Kurukshetra.